FINAL YEAR, VII SEMESTER

Crop Process Engineering (EAG-701)

UNIT- I

Scope and importance of food processing, principles and methods of food processing. Processing of farm crops; cereals, pulses, oil seeds, fruits and vegetables and their products for food and feed. Processing of animal products,

UNIT- II

Principle of size reduction, grain shape, Size reduction machines; crushers, grinders, cutting machines etc. - operation, efficiency and power requirement – Rittinger's, Kick's and Bond's equation, fineness modulus.

UNIT- III

Theory of mixing, types of mixtures for dry and paste. materials, rate of mixing and power requirement, mixing index. Theory of separation, size and unsized separation,

UNIT- IV

Types of separators, size of screens, sieve analysis, capacity and effectiveness of screens, pneumatic separation. Theory of filtration, study of different types of filters, rate of filtration, pressure drop during filtration.

UNIT- V

Scope & importance of material handling devices, study of different types of material handling systems; belt, chain and screw conveyor, bucket elevator, pneumatic conveying, gravity conveyor- design consideration, capacity and power requirement.

Crop Process Engineering Lab (EAG-751)

1. Preparation of flow and layout charts of a food processing plant
2. Determination of fineness modulus and uniformity index
3. Performance evaluation of hammer mill
4. Performance evaluation of attrition mill
5. Study of cleaning equipment
6. Separation behaviour in pneumatic separation
7. Study of grading equipment
8. Evaluation of performance of indented cylinder and screen pre-cleaner
9. Mixing index and study of mixers
10. Study of conveying equipments
11. Performance evaluation of bucket elevator.

Soil and Water Conservation Structures (EAG-702)

UNIT- I

Introduction; classification of structures, functional requirements of soil erosion control structures; flow in open channels-types of flow, state of flow, regimes of flow, energy and momentum principles, specific energy and specific force; hydraulic jump and its application, type of hydraulic jump, energy dissipation due to jump, jump efficiency, relative loss of energy;

UNIT- II

Runoff measuring structures-parshall flume, H - flume and weirs; straight drop spillway - general description, functional use, advantages and disadvantages, structural parts and functions; components of spillway, hydrologic and hydraulic design, free board and wave free board, aeration of weirs, concept of free and submerged flow,

UNIT- III

Structural design of a drop spillway-loads on headwall, variables affecting equivalent fluid pressure, determination of saturation line for different flow conditions, seepage under the structure, equivalent fluid pressure of triangular load diagram for various flow conditions, creep line theory, uplift pressure estimation, safety against sliding, over turning, crushing and tension;

UNIT- IV

Chute spillway general description and its components, hydraulic design, energy dissipaters, design criteria of a SAF stilling basin and its limitations,

UNIT- V

Drop inlet spillway- general description, functional use, design criteria; design of diversions; small earth embankments-their types and design principles, farm ponds and reservoirs, cost estimation of structures.

FINAL YEAR, VIII SEMESTER

Drying and Storage Engineering (EAG-801)

UNIT- I

Moisture content and methods for determination, importance of EMC and methods of its determination, EMC curve and EMC model, principle of drying, theory of diffusion, mechanism of drying- falling rate, constant rate, thin layer, deep bed and their analysis, critical moisture content, drying models.

UNIT- II

Calculation of drying air temperature and air flow rate, air pressure within the grain bed, Shred's and Hukill's curve, different methods of drying including puff drying, foam mat drying, freeze drying, etc. Study of different types of dryers- performance, energy utilization pattern and efficiency, study of drying and dehydration of agricultural products.

UNIT- III

Types and causes of spoilage in storage, conditions for storage of perishable products, functional requirements of storage, control of temperature and relative humidities inside storage, calculation of refrigeration load; modified atmospheric storage and control of its environment, air movement inside the storage,

UNIT- IV

Storage of grains: destructive agents, respiration of grains, moisture and temperature changes in stored grains; conditioning of environment inside storage through natural ventilation,

mechanical ventilation, artificial drying, grain storage structures such as Bukhari, Morai, Kothar, silo, CAP, warehouse - design and control of environment.

UNIT- V

Storage of cereal grains and their products, storage of seeds, hermetically sealed and air-cooled storages-refrigerated, controlled atmosphere, modified atmospheric and frozen storages. Storage condition for various fruits and vegetables under cold and CA storage system. Economic, aspects of storage.

Agricultural Structures and Environmental Control (EAG-802)

UNIT- I

Planning and layout of farmstead. Physiological reactions of livestock to solar radiation and other environmental factors, livestock production facilities, BIS. Standards for dairy, piggery, poultry and other farm structures. Design, construction and cost estimation of farm structures; animal shelters, compost pit, fodder silo, fencing and implement sheds, barn for cows, buffalo, poultry, etc.

UNIT- II

Design and construction of rural grain storage system engineering for rural living and development, rural roads, their construction cost and repair and maintenance. Sources of water supply, norms of water supply for human being and animals, drinking water standards and water treatment suitable to rural community.

UNIT- III

Site and orientation of building in regard to sanitation, community sanitation system; sewage system and its design, cost and maintenance, design of septic tank for small family.

UNIT- IV

Estimation of power requirement for domestic and irrigation, source of power supply, use of alternate source of energy, electrification of rural housing. Scope, importance and need for environmental control, renewable and non-renewable resources and their equitable use.

UNIT- V

Concept of eco system, biodiversity of its conservation, environmental pollution and their control, solid waste management system, BOD and COD of food plant waste, primary and secondary treatment of food plant waste

ELECTIVE I

Rural Water Supply and Sanitation, (EAG-011)

UNIT- I
Introduction and scope of the subject, types of roads planning of rural roads, geometric design of roads,
UNIT- II
pavement design, material for construction of roads, source and characteristics of water supply,
UNIT- III
estimation of quantity for rural water supply, layout of distribution system intake works,
UNIT- IV
capacity requirement of storage of water, water treatment methods, pipe network analysis,
UNIT- V
estimation and design of sewage systems, septic tanks.

Rural Engineering, (EAG-012)

UNIT- I
Planning & design of rural roads, design and construction of culverts,
UNIT- II
design and planning of community development buildings, planning and design of residential and non-residential buildings,
UNIT- III
planning and design of stadium, planning and design of fish ponds,
UNIT- IV
flood protection works, rural water supply system, pond/tank water distribution and planning
UNIT- I
design of rural drainage.

Watershed Planning and Management (EAG-013)

UNIT- I
Watershed management - problems and prospects; watershed based land use planning, watershed characteristics – physical and geomorphologic, factors affecting watershed management, hydrologic data for watershed planning,
UNIT- II
Watershed delineation, delineation of priority watershed, water yield assessment and measurement from a watershed; hydrologic and hydraulic design of earthen embankments and diversion structures;
UNIT- III
Sediment yield estimation and measurement from a watershed and sediment yield models; rainwater conservation technologies - in-situ and storage, design of water harvesting tanks and ponds; water budgeting in a watershed; effect of cropping system,

UNIT- IV

Land management and cultural practices on watershed hydrology; evaluation and monitoring of watershed programmes; people's participation in watershed management programmes;

UNIT- V

Planning and formulation of project proposal; cost benefits analysis of watershed programmes; optimal land use models; case studies.

Waste and By-Product Utilization, (EAG-014)

UNIT- I

Types and formation of by products and waste; magnitude of waste generation in different food processing industries; concept scope and maintenance of waste management and effluent treatment, Temperature, pH, Oxygen demands (BOD, COD), fat, oil and grease content, metal content, forms of phosphorous and sulphur in waste waters, microbiology of waste, other ingredients like insecticide, pesticides and fungicides residues,

UNIT- II

Waste utilization in various industries, furnaces and boilers run on agricultural wastes and by products, briquetting of biomass as fuel, production of charcoal briquette, generation of electricity using surplus biomass, producer gas generation and utilization,

UNIT- III

Waste treatment and disposal, design, construction, operation and management of institutional community and family size biogas plants, concept of vermi-composting, Pre-treatment of waste: sedimentation, coagulation, flocculation and floatation,

UNIT- IV

Secondary treatments: Biological and chemical oxygen demand for different food plant waste– trickling filters, oxidation ditches, activated sludge process, rotating biological contractors, lagoons, Tertiary treatments:

UNIT- V

Advanced waste water treatment process-sand, coal and activated carbon filters, phosphorous, sulphur, nitrogen and heavy metals removal, Assessment, treatment and disposal of solid waste; and biogas generation.

Bioconversion Engineering, (EAG-015)

UNIT- I

Introduction, definition, importance and justification of bioconversion and bioconversion engineering; Importance groups of microorganisms and enzymes, their structures and functions;

UNIT- II

Growth of microorganisms, different phases, growth kinetics, methods of measurement, effect of fermentation conditions, design of growth media;

UNIT- III

Description of conventional bioconversion process for production of commodity items (single cell protein, antibiotics, polysaccharide gums, organic acids, solvents etc.); Different types of reactors (batch, feed batch continuous),

UNIT- IV

Reactor design, process control, media sterilization; Downstream processing for recovery of fermentation products (filtration, centrifugation and other processes), instrumentation for bioconversion processes.

UNIT- V

Bioconversion of agricultural wastes (lignocellulosic), solid state fermentation, fermented foods and feeds, anaerobic digestion of municipal and agricultural wastes.

Food Processing Plant Design & Layout, (EAG-016)

UNIT- I

Meaning and definition of plant layout. Objectives and principles of layout. Types of layout.

UNIT- II

Salient features of processing plants for cereals, pulses oilseeds, horticultural and vegetable crops, poultry, fish and meat products, milk and milk products.

UNIT- III

Location selection criteria, selection of processes, plant capacity, project design, flow diagrams,

UNIT- IV

Selection of equipments, process and controls, handling equipments, plant layout, Plant elevation, requirement of plant building and its components, labour requirement,

UNIT- V

Plant installation, power and power transmission, sanitation. Cost analysis, preparation of feasibility report.

Ergonomics Principles and Practices, (EAG-017)

UNIT- I

Human factors in system development – concept of systems; basic processes in system development, performance reliability, human performance.

UNIT- II

Information input process, visual displays, major types and use of displays, auditory and factual displays. Speech communications. Biomechanics of motion, types of movements,

UNIT- III

Range of movements, strength and endurance, speed and accuracy, human control of systems. Human motor activities, controls, tools and related devices.

UNIT- IV

Anthropometry: arrangement and utilization of work space, atmospheric conditions, heat exchange process and performance, air pollution. Dangerous machine (Regulation) act,

UNIT- V

Rehabilitation and compensation to accident victims, Safety gadgets for spraying, threshing, Chaff cutting and tractor & trailer operation etc.

Land Development Machinery, (EAG-018)

UNIT- I

Engineering fundamentals related to earth moving machinery; terminology and nomenclature, swell, shrinkage, rolling resistance, reactive effort, rim pull etc. Grade resistance and

gradability. Soil strength parameters, soil stabilization and compaction equipment. Tractors and crawlers; classification, working principle and capacity determination, motorgrader and land levelling equipment: levellers and land plane.

UNIT- II

Land clearing equipment; types and application. Front end loaders; Land excavating equipment; scrapers, power shovel, drag line, clamshell, hoe and trenching machinery: Trucks and wagons; types, capacities and performance. Engg. Fundamentals of moving the earth-rolling, resistance. The effect of grade on required tractive effort, coefficient of traction. Effect of altitude on performance of I.C.engine. Effect of temperature on the performance of I.C. engines. Combined effect of pressure and temperature on the performance of I.C.engines. Drawbar pull.

UNIT- III

Tractors and Related Equipment: Tractor uses, types of tractor crawler tractors; Bulldozers: General information, Crawler mounted versus wheel mounted bulldozers, Moving earth with bulldozers, output of bulldozers. Land Clearing Operations: Types of equipment used. Scrapers: General information, types and sizes, Performance of wheel type scrapers. Excavating Equipment: Power shovel; general information, size of power shovel, basic parts, working principle.

UNIT- IV

Draglines-General Information, types of draglines, size, basic parts, optimum depth, output, effect of depth of cut and angle of swing on output. Clamshell: Types of bucket, sizes, production rates for clamshells. Hoes: General information, basic parts of cable operated hoe. Working ranges of hoes, output of hoes. Trenching Machines: General Information, wheel type trenching machine. Ladder type trenching machine. Trucks and Wagons: Trucks, classification and utilization.

UNIT- V

Types, output performance capabilities of trucks and wagons. Balancing capacities of hauling units with the size of excavator. Effect of the size of truck on the cost of hauling unit. Front end loaders output; levellers, output and operation.

Mechanics of Tillage and Traction, (EAG-019)

UNIT- I

Introduction to mechanics of tillage tools, engineering properties of soil, principles and concepts, stress strain relationship,

UNIT- II

design of tillage tools principles of soil cutting, design equation, force analysis, application of dimensional analysis in soil dynamics performance of tillage tools.

UNIT- III

Introduction to traction and mechanics, off road traction and mobility, traction model, traction improvement, traction prediction,

UNIT- IV

tyre size, tyre lug geometry and their effects, tyre testing, soil compaction and plant growth,

UNIT- V

variability and geo statistic, application of GIS in soil dynamics.

ELECTIVE II

Minor Irrigation and Command Area Development, (EAG-021)

UNIT- I
Major, medium and minor irrigation projects – their comparative performance; development and utilization of water resources through different minor irrigation schemes.
UNIT- II
Basic concepts of command area – definition, need, scope, and development approaches: historical perspective, command area development authorities;
UNIT- III
Interaction/collaboration of irrigation water use efficiency and agricultural production.
UNIT- IV
Planning and execution of on farm development activities with in the scope of command area development; Use of remote sensing techniques for command area development;
UNIT- V
Case studies of some selected commands; Farmers participation in command area development.

Aquacultural Engineering, (EAG-022)

UNIT- I
Land survey, area calculation of plane surface of regular and irregular shape as applied to measurement of land, trapezoidal rule, Simpson's rule, volume of regular and irregular shape as applied to the volume of stacks, sheds, heaps. Farm-types and objectives; Fresh water and coastal aquafarms.
UNIT- II
Preliminary survey, site selection, topography. Land survey – chain surveying, compass surveying, leveling, plane table surveying and contour surveying; Soil – types, properties, classification, sampling methods and texture analysis.
UNIT- III
Location, design and construction of hatcheries, race ways and farm complex. Tide-fed / pump fed farms, creeks, estuarine and marine water source utilization. Open canals and their types. Sluices and gates.
UNIT- IV
Earth work calculation – ponds, dykes, canals and roads. Design and construction of ponds and dykes. Tidal influences and maintenance; Effect of seepage and evaporation and their control. Water budgeting. Water distribution system – main feeder channel, drainage channel.
UNIT- V
Water control structure – types of inlets and outlet and their construction. Computation of water intake and discharge. Aerators – principles, classification and placement. Pumps- types, total head and horse power. Filters – types and construction.

Reinforced Cement Concrete, (EAG-023)

Unit I

Reinforced Cement Concrete, Concept of Composite material, Purpose of providing reinforcement, materials used in R.C.C and their requirements, different grades of cement and steel, Characteristic strength and grades of concrete, modular ratio of R.C.C, types of loads on structures,

Unit II

Design of singly reinforced T-beams–cantilever/simply supported beams, Design of continuous beams, Design of Lintel simple problems, Classification of slabs, Effective spans, Imposed loads on slabs, strength and stiffness requirements, minimum and maximum permitted size, spacing and area of main and secondary reinforcement,

Unit III

Design of cantilever, simply supported, slabs and sunshades by limit state method, Design of continuous slabs using B.M coefficients, check for stiffness, curtailment of tension reinforcement

Unit IV

Limit state of collapse in compression, assumptions, limiting strength of short axially loaded compression members, effective length of compression members, slenderness limits for columns, classification of column-minimum eccentricity for column loads, longitudinal and transverse reinforcement, Design of axially loaded short columns with lateral ties / helical reinforcement

Unit V

Classification of beams, lateral buckling of beams, assumptions, permissible bending and shear stresses, minimum thickness of elements, limiting deflection of beams, Design of laterally supported beams.

Gulley and Ravine Control Structures (EAG-024)

UNIT- I

Introduction; floods - causes of occurrence, flood classification - probable maximum flood, standard project flood, design flood, flood estimation - methods of estimation;

UNIT- II

Estimation of flood peak - Rational method, empirical methods, Unit hydrograph method; Statistics in hydrology, flood frequency methods - Log normal,

UNIT- III

Gumbel’s extreme value, Log-Pearson type-III distribution; depth-area-duration analysis; flood forecasting, flood routing – channel routing, Muskingum method, reservoir routing, modified Pul’s method;

UNIT- IV

Flood control - history of flood control, structural and non-structural methods of flood control measures, storage and detention reservoirs, levees, channel improvement;

UNIT- V

Gulley erosion and its control; soil erosion and sediment control measures; river training works, planning of flood control projects and their economics.

Development of Processed Products & Equipments (EAG-025)

UNIT- I

Applications of unit operations to the food industry, analytical processing concepts with regards to mass and energy balances, equipment involved in the commercially important food processing methods and unit operations; value addition to cereals like rice, wheat etc.

UNIT- II

Parboiling of rice, quality of processed products of rice & wheat. Processing of pulses, spices and condiments; extruded food product, fermented food product, frozen and dried product, technology of meat, fish and poultry products,

UNIT- III

Technology of milk and milk products. Technology of oilseeds and fat products, snack foods, Fruits and vegetables product: candy, nutraceuticals, food product development trends, food additives and labeling.

UNIT- IV

Process equipment for thermal processing-evaporation, dehydration, drying, blenching, pasteurization, distillation; mechanical separation-filtration, sieving, centrifugation, sedimentation;

UNIT- V

Mechanical handling-conveying and elevation; size reduction and classification-mixing; kneading, blending.

Medicinal and Aromatic Plants, Production and Processing, (EAG-026)

UNIT- I

History, scope, opportunities and constraints in the cultivation and utilization of medicinal and aromatic plants in India. Importance, origin, distribution, area, production, climatic and soil requirements,

UNIT- II

Propagation and nursery techniques, planting and aftercare, training and pruning, nutritional and water requirements. Plant protection, harvesting, processing and economics of under mentioned important medicinal and aromatic plants.

UNIT- III

Medicinal Plants : pepper, cardamom, clove, ginger, turmeric, betelvine, periwinkle, Rauvolfia, Dioscorea, isabgol, Ammi majus, belladonna, Cinchona, pyrethrum and other species relevant to local conditions.

UNIT- IV

Aromatic Plants : Citronella grass, khus grass, sweet flag (bach), lavender, geranium, patchouli, bursera, Mentha, muskdana (musk mallow), Ocimum and other species relevant to the local conditions. Endangered medicinal and aromatic plants of India and their conservation.

UNIT- V

Study of chemical composition of a few important medicinal and aromatic plants, their extraction and use. Therapeutic and pharmaceutical uses of important species.

Biomass Management for Fodder and Energy, (EAG-027)

UNIT- I

Introduction to biomass management, biomass resource assessment management techniques/supply chains,

UNIT- II

Processing of paddy straw, densification- Extrusion process, pellets, mills and cubers, Bailing-classification, uses;

UNIT- III

Residue management for surface mulch and soil incorporation, Paddy Straw choppers and spreaders as an attachment to combine Harvester,

UNIT- IV

Mulch seeder, Paddy Straw Chopper-cum-Loader, Balar for collection of straw; Processing of straw/ fodder for animal use; Agricultural and horticultural use,

UNIT- V

Cushioning material for fruits and vegetables, Mulching and Composting, Paper and cardboard manufacturing, Straw as a fuel.

Production Technology of Agricultural Machinery, (EAG-028)

UNIT- I

Critical appraisal in production of Agricultural Machinery; Modelling and stress analysis of Machinery parts by using standard software;

UNIT- II

Advances in material used for tractor & Agril. Machinery. Cutting tools including CNC tools and finishing tools. Advanced manufacturing techniques like powder metallurgy, EDM (Electro-Discharge Machining),

UNIT- III

Heat Treatment of steels including pack carburizing, shot pining process, chemical vapour deposition (CVD) etc. Limits, Fits & Tolerances, Jigs & Fixtures, Microstructure Analysis.

UNIT- IV

Industrial lay-out planning, Quality management,. Economics of process selection. Techno-Economic feasibility of Project Report. Selection of Standard/ critical components. Case studies of manufacturing of agril. machinery.

UNIT- V

Servo motors, drives & controllers, CNC controllers for machine tools. CNC programming. Assembly and plant automation. Storage and transportation.

Fuels, Lubricants and Maintenance (EAG-029)

Unit-I

Engine friction: introduction, total engine friction, effect of engine variables on friction, hydrodynamic lubrication, elasto hydrodynamic lubrication, boundary lubrication, bearing lubrication, functions of the lubrication system, introduction to design of a lubricating system.

Unit-II

Specific requirements for automotive lubricants, oxidation deterioration and degradation of lubricants, additives and additive mechanism, synthetic lubricants, classification of lubricating oils, properties of lubricating oils, tests on lubricants. Grease, classification, properties, test used in grease.

Unit-III

Thermo-chemistry of fuels, properties and testing of fuels, relative density, calorific value, distillation, vapour pressure, flash point, spontaneous ignition temperature, viscosity, pour point, flammability, ignitability, diesel index, API gravity, aniline point etc.

Unit-IV

SI Engines, flame propagation and mechanism of combustion, normal combustion, knocking, octane rating, fuel requirements.

Unit-V

CI Engine, mechanism of combustion, diesel knock, cetane rating, fuel requirements. Additive, mechanism, requirements of an additive, petrol fuel additives and diesel fuel additives, specifications of fuels.

ELECTIVE III

Reservoir and Farm Pond Design (EAG-031)

UNIT- I

Earthen embankments - functions, advantages and disadvantages, classification – hydraulic fill and rolled fill dams - homogeneous, zoned and diaphragm type;

UNIT- II

Foundation requirements, grouting, seepage through dams - estimation of seepage discharge, location of seepage/phreatic line by graphical and analytical methods,

UNIT- III

Flow-net and its properties, seepage pressure, seepage line in composite earth embankments, drainage filters, piping and its causes;

UNIT- IV

Design and construction of earthen dam, stability of earthen embankments against failure by tension, overturning, sliding etc; stability of slopes - analysis of failure by slice method;

UNIT- V

Types of reservoirs and farm ponds, design and estimation of earth work; cost analysis.

Remote Sensing and GIS Application (EAG-032)

UNIT- I

Remote Sensing: Definition, stage in remote sensing, modern remote sensing technology versus conventional aerial photography; visual image interpretation, image interpretation,

UNIT- II

Basic principles of image interpretation, factors governing the quality of an image; factors governing interpretability, visibility of objects, elements of image interpretation, techniques of image interpretation,

UNIT- III

Digital image processing, digital image; remote sensing in agriculture progress and prospects, microwave radiometry for monitoring agriculture crops and hydrologic forecasting; aerial photo interpretation for water resources development and soil conservation survey.

UNIT- IV

GIS: History of development of GIS definition, basic components, and standard GIS packages; data-entry, storage and maintenance; data types-spatial-non-spatial (attribute data),data structure,

UNIT- V

Data format- point line vector-raster – polygon-object structural model, files, files organization-data base management systems (DBMS), entering data in computerdigitizer-scanner-data compression.

Micro Irrigation Systems Design, (EAG-033)

UNIT- I

Past, present and future need of micro-irrigation systems, Role of Govt. for the promotion of micro-irrigation in India, Merits and demerits of micro-irrigation system,

UNIT- II

Types and components of micro-irrigation system, Micro-irrigation system- design, design synthesis, installation, and maintenance. Sprinkler irrigation - types, planning factors, uniformity and efficiency, laying pipeline, hydraulic lateral, sub-mains and main line design,

UNIT- III

Pump and power unit selection. Drip irrigation – potential, automation, crops suitability. Fertigation – Fertilizer application criteria, suitability of fertilizer compounds, fertilizer mixing, injection duration, rate and frequency, capacity of fertilizer tank.

UNIT- IV

Quality control in micro-irrigation components, design and maintenance of polyhouse; prospects,

UNIT- V

Waste land development – hills, semi-arid, coastal areas, water scarce areas, Benefit and Cost analysis. Factors affecting shelf life of food material during storage; spoilage mechanism during storage;

Seed Processing, (EAG-034)

UNIT- I

Seed processing and its importance, principles of seed processing, seed industry and seed act; Development of seed industry in India, Preparing seed for processing; Seed conditioning machines. Debearder machines, Hand and power operated shelling machines; Seed cleaning machine;

UNIT- II

Seed drying, natural and mechanical, dryers: their types & operation and maintenance; Seed cleaning; different type of air-screen machines, operation and maintenance; Seed grading; different types of seed graders such as length and breadth separators, disc separators, Indented cylinder separator, gravity separator; Stoner; Air classifier Magnetic separators, colour separators, etc.

UNIT- III

Seed treatment; types of treatment, methods and related equipments such as liquid treaters, slurry treaters, dust and fumigants, precautious regarding the seed treatment and ISI recommendations; Seed packaging, stiching and bag closing machines, automatic weighing machines and tagging;

UNIT- IV

Seed storage; principles of seed storage, storage structures; Dehumidifiers to control temp. and moisture, changes in seed quality during storage. Cap storage of hermetically sealed storages; Grain bins and silos, drying-cum-storage bins;

UNIT- V

Seed conveyors; bucket elevators, belt conveyors, screw conveyors, trucks bagons; Repair and maintenance of different types of conveying devices; Seed plant layout and construction

Food Packaging Technology, (EAG-035)

UNIT- I

Definition, requirement, importance and scope of packaging of foods; types and classification of packaging system; advantage of modern packaging system. Different types of packaging materials used.

UNIT- II

Different forms of packaging, metal container, glass container, plastic container, flexible films, shrink packaging, vacuum & gas packaging. Packaging requirement & their selection for the raw & processed foods.

UNIT- III

Advantages & disadvantages of these packaging materials; effect of these materials on packed commodities,

UNIT- IV

Package testing, Printing, labeling and lamination. Economics of packaging; performance evaluation of different methods of packaging food products; their merits and demerits;

UNIT- V

Scope for improvements; disposal and recycle of packaging waste.

Design and Maintenance of Greenhouse, (EAG-036)

UNIT- I

History and types of greenhouse; importance, function and features of green house; scope and development of green house technology. Location, Planning and various component of greenhouse;

UNIT- II

Design criteria and calculation; constructional material and methods of construction; covering materials and its characteristics, solar heat transfer, solar fraction for green house, steady state analysis of green house,

UNIT- III

Greenhouse heating, cooling, shedding and ventilation systems; Carbon Dioxide generation and monitoring and lighting systems, instrumentation & computerized environmental Control Systems.

UNIT- IV

Watering, fertilization, root substrate and its pasteurization, containers and benches, plant nutrition. Alternative cropping systems; plant tissue culture, chemical growth regulation; disease control; integrated pest management;

UNIT- V

Postproduction quality and handling Cost analysis of greenhouse production; Applications of green house & its repair & maintenance.

Testing of Agricultural Machinery and Tractors, (EAG-037)

UNIT- I

Basic objectives of testing of agricultural equipment; concept and importance. Different terminology used in testing of agricultural equipment, I.C engines and tractors. Different types of transducer and their uses, instrumentation for draft, slip, sinkage, soil resistance, sound and vibration. Hardness of material. Test codes: Nebraska, OECD, RNAM,BS,ISO,BIS etc. Testing of primary and secondary tillage tools, sowing/planting equipment, weeding tools,

UNIT- II

Harvesting machinery, power threshers, combine harvesters, sprayers, dusters etc. as per BIS standard. Testing of I.C. engines and tractors as per BIS test codes. Basic objectives of testing agricultural equipment: Concept and importance. Different terminology used in testing of agricultural equipment, I.C. engines and tractors. Different types of transducers and their uses, instrumentation for draft, slip, sinkage, soil resistance, sound and vibration measuring instruments. Hardness measuring instruments.

UNIT- III

Test codes: Nebraska, OECD, RNAM, BS, ISO, BIS etc., basis difference. Test codes and procedure for primary tillage tool as per BIS test code. Test code and procedure for testing of secondary as per BIS test code. Test code and procedure for testing of ferti-seed drill as per BIS test code.

UNIT- IV

Test code and procedure for testing of weeding tools as per BIS test code. Test code and procedure for testing of a harvesting machine as per BIS test code. Test code and procedure for testing of a power thresher as per BIS test code. Test code and procedure for testing of a combine harvester as per BIS test code.

UNIT- V

Test procedure of a power sprayer as per BIS test code. Test code and procedure of a agricultural tractor as per BIS test code. Test code and procedure of an I.C engine as per BIS test code.

Tractor Design Principles, (EAG-038)

UNIT- I

Procedure for design and development of agricultural tractor, Study of parameters for balanced design of tractor for stability & weight distribution,

UNIT- II

Hydraulic lift and hitch system design. Design of mechanical power transmission in agricultural tractors.

UNIT- III

Design of Ackerman Steering and tractor hydraulic systems.

UNIT- IV

Study of special design features of tractor engines and their selection.

UNIT- V

Design of seat and controls of an agricultural tractor. Tractor Testing.